

SHAKYA BHATTACHARYYA

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SUMMARY

Business-focused Data Scientist and Decision Systems Builder with 5+ years of experience designing and deploying end-to-end analytics, ML, and AI-driven systems. Proven track record of translating ambiguous operational and financial problems into production-ready data pipelines, predictive models, and decision frameworks influencing \$30M+ in cost and revenue outcomes. Strong in Python, SQL, and applied ML with growing specialization in **LLM-integrated systems, real-time event pipelines, and operational intelligence platforms**. Experienced in owning systems from ingestion to decision delivery, aligning closely with fintech, risk, and operations analytics domains.

SKILLS

- **Programming & ML:** Python (pandas, numpy, scikit-learn, TensorFlow, XGBoost), SQL, Git
- **AI & NLP:** LLM Systems, Prompt Engineering, BERT, TF-IDF, Text Classification, Image Recognition
- **Analytics & Modeling:** Predictive Modeling, Forecasting, A/B Testing, Credit Risk Modeling, Time-series, Clustering, Uplift Modeling
- **Data Engineering:** ETL, Data Cleaning, Feature Engineering, Batch Scoring, API Deployment, Event Processing, API Design, PostgreSQL
- **System & Tools:** Docker, MySQL, FastAPI, Docker, AWS S3, Azure, Postman, Cursor
- **BI & Reporting:** Tableau, PowerBI, Executive Dashboards, KPI Design

PROFESSIONAL EXPERIENCE

Rock Emergency Services

Feb 2026 - Present

Senior Data Scientist - AI Systems and Decision Intelligence

Rochester, NY

- Owned the end-to-end architecture and development of an event-driven operational intelligence platform that ingests real-time data via webhooks and APIs, detects issues, triggers alerts, and automates workflows, resulting in faster incident response across business operations
- Designed a **3-layer data architecture (raw → normalized mirror → canonical)** enabling scalable multi-source ingestion, auditability, and downstream decision systems.
- Developed a tooling layer and query interfaces that retrieve structured operational data on projects, financials, and staffing, enabling faster decision-making and automation of routine tasks
- Implemented LLM-integrated agent workflows that translate natural-language queries into system-level actions and insights, streamlining information access during early-stage deployment
- Built **end-to-end pipeline (FastAPI + PostgreSQL + webhook ingestion + Slack delivery)** enabling real-time monitoring and alerting across operational events.

Simon Business School

Jan 2025 - Dec 2025

Instructional Analytics Associate - Analytics Design & Application

Rochester, NY

- Led analytics labs for 100+ graduate students, teaching production-style workflows (SQL, Python, ML pipelines, BI delivery), which helped students complete projects independently and improve their practical analytics skills
- Designed end-to-end case systems covering ETL → modeling → decision dashboards, mirroring real-world analytics implementations, enabling students to apply the full analytics cycle in coursework and deepen their hands-on experience
- Built automated grading and feedback systems using Python, reducing evaluation time by **70% and improving consistency**.

Simon Vision Consulting

Sep 2024 - Dec 2024

Data Science Analyst (Client Engagement: City of Rochester)

Rochester, NY

- Built retention and classification models improving AUC by **0.07**, identifying key drivers of student retention and informing intervention strategies.
- Designed and executed A/B tests and uplift models in Python (pandas, statsmodels) to evaluate program effectiveness, providing evidence that guided more efficient resource allocation across initiatives
- Developed **Power BI dashboards integrating SQL + Excel pipelines**, reducing reporting effort by **50%** and improving executive decision visibility.
- Applied NLP (TF-IDF + topic modeling) on **5,000+ survey responses**, extracting actionable themes that shaped policy decisions.

Cognizant Technology Solutions

Feb 2022 - Jun 2024

Business Data Analyst (ML Focus)

Kolkata, India

- Built ML-based forecasting and risk models (XGBoost, regression) on **250K+ clinical trial records**, reducing decision turnaround time by **20%** and supporting **\$30M+ cost avoidance**.
- Translated model outputs into operational decision frameworks using Python dashboards and SQL queries, enabling faster resource-allocation decisions and more effective risk prioritization
- Designed automated **ETL + batch scoring pipelines (SQL + scheduling)** reducing reporting cycles by **~90%**.
- Deployed TensorFlow-based models improving detection precision by **12%**, directly impacting operational triage accuracy.
- Improved feature engineering pipelines, reducing latency and increasing efficiency by **20% across production workflows**.

PROJECTS

Fraud Operations Analytics & Risk Decisioning | [Project Link](#)

Aug 2025 - Dec 2025

- Designed risk scoring and prioritization system aligning fraud signals with **business cost thresholds and operational constraints**.
- Reduced false positives by **25–35%** and manual review workload by **~20%**.
- Identified gaps in existing decision logic and improved targeting precision across fraud workflows.

EDUCATION

Simon Business School, University of Rochester

Jul 2024 - Dec 2025

Master of Science, Business Analytics (STEM)

Rochester, NY

- **GPA:** 3.9/4.0
- **Achievements:** Advanced Certificate in Pricing Analytics (Revenue Optimization Focus)

Maulana Abul Kalam Azad University of Technology

Aug 2018 - Jun 2022

Bachelor of Technology, Computer Science & Engineering (STEM)

India